

ENRIQUE FRANCISCO ALEMANY MARTINEZ

Senior Graphics / Core Programmer | 46980, Valencia Spain | +34 680 769 630 | enrique.alemany@pm.me | [LinkedIn](#)

SUMMARY

Graphics Programmer with over 7 years of experience in real-time rendering and shader development. I specialize in low-level optimization across DirectX and Vulkan, with a focus on solving complex GPU bottlenecks and improving engine performance. My background includes building artist-facing tools, delivering rendering features, developing graphics profiling and debugging tools, and extending/modifying modern Game Engines.

SHIPPED PROJECTS

Mafia: The Old Country | Spectre Divide | Total War: Warhammer III | Total War: Three Kingdoms | A Total War Saga: Troy

SKILLS

Engines: Unreal Engine, Unity, Warscape (Creative Assembly)
APIs & Languages: DirectX 11, DirectX 12, Vulkan, HLSL, GLSL, C++
Tools: Nsight, Renderdoc, Pix, Razor
Expertise: GPU profiling, Shader optimization, In-Engine Tools, Cross-team collaboration

EXPERIENCE

Senior Core & Rendering Engineer

11/2025 – Present

31st Union · Valencia, Spain (Hybrid)

- Investigated and profiled Nanite performance bottlenecks, delivering actionable insights to optimize frame time and GPU throughput.
- Designed and implemented Unreal Engine 5 PSO hitch tooling and fixes, automating the PSO bundle generation process.
- Eliminated manual bundle generation and the need to author custom level updates, saving days of development time.
- Provided cross-team rendering support, fixed core Unreal crashes, managed engine integrations, and created technical VFX guidelines.

Senior Graphics Programmer (6-Month Contract)

04/2025 – 09/2025

CodeDev (The Unreal Guys) · Valencia, Spain

- Delivered performance optimization support in Unreal Engine 5, focusing on Nanite, VSM, and Lumen systems.
- Conducted deep-dive GPU performance analysis for PC, Xbox Series X/S, and PS5, identifying and resolving platform-specific hotspots.
- Integrated Nvidia Aftermath for advanced crash debugging and resolved general GPU stability issues.
- Managed core Unreal Engine 5.4+ source-level integrations.

Mountaintops studios (Studio closure due to lack of funding)

08/2022 – 04/2025

Valencia, Spain

Senior Graphics & Tools Engineer

02/2025 – 04/2025

- Optimized compute shaders for a proprietary, custom ink rendering pipeline.
- Spearheaded console optimizations (PS5, Xbox Series X/S) and re-implemented rendering features to satisfy min-spec PC requirements.
- Collaborated directly with technical artists to optimize shader workflows and engine subsystems.

Graphics & Tools Engineer

08/2022 – 02/2025

- Engineered a custom Blender LiveLink tool to communicate seamlessly with internal inking tools, enabling real-time asset editing inside Unreal Engine.
- Improved the ink compute shader pipeline performance by 75% by re-architecting it into a fully GPU-driven implementation.
- Reduced the overall memory footprint of the ink rendering system while introducing new creative features for artists.

Creative assembly

06/2019 – 08/2022

Horsham, UK

Graphics Programmer

04/2021 – 08/2022

- Focused on GPGPU programming, VFX system expansions, and core rendering pipeline enhancements.
- Built a fully GPU-driven ribbon rendering system from scratch for the Warscape engine, enabling dynamic ribbons with custom materials at minimal GPU cost.
- Partnered with the VFX team to implement debugging features and improve tool usability.

Associate Graphics Programmer

04/2020 – 04/2021

- Maintained core VFX systems and implemented modern real-time VFX rendering techniques to enhance overall engine performance.

Trainee Graphics Programmer

06/2019 – 04/2020

- Gained hands-on engine architecture experience, collaborating across disciplines to execute on creative and visual targets.

EDUCATION

BSc Computer Science For Games (Top Mark)

06/2019

Sheffield Hallam University · UK

- Final Year Project: Developed a custom Monte Carlo path tracer built using OpenGL compute shaders and C++.

HND Computing & Systems Development For Games

06/2018

ESAT · Valencia, Spain